Instructions:

- Solutions to the problem sets are due by 9:00 a.m. the next morning.
- While points may differ between Problem Sets each Problem Set will carry equal weight for course grade.
- Please list your collaborators on your submission.

Problem 1. Particle Velocities (20 pts)

(a) (5 pts) An electron is accelerated between the plates of a parallel plate capacitor biased to potential V starting form rest at the minus electrode (see Figure 1). Give formulas for the final speed v of the electron in the nonrelativistic approximation and relativistically.

(b) (5 pts) Using your answer for (a), plug in values V = 10V and V = 10 kV for both the nonrelativistic and relativistic formulas. Express answers in m/sec and $\beta = v/c$ units.

(c) (5 pts) Protons in medical accelerators have kinetic energy $\sim$ 100 MeV. Estimate γ, β, and the speed (in m/sec) of the protons.

(d) (5 pts) In heavy-ion accelerators ion kinetic energies are typically measured in MeV/nucleon. Show how this approximately fixes the particle axial β. Explain why this is desirable in resonant RF accelerators.

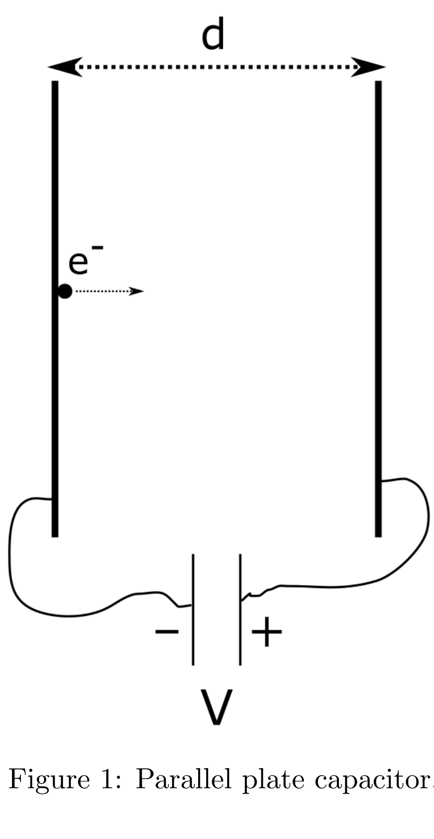

Figure 1: Parallel plate capacitor.

Problem 2. RHIC frequency/field (10 pts)

The RHIC collider collides fully stripped gold ions (A=197, Z=79) at a total energy of E_{coll}=100 GeV/nucleon per beam. The circumference of each ring is 3834 m. Assume the mass of a gold ion is 197×0.93113 GeV/c^2.

(a) (5 points) Calculate the revolution frequency of a particle at the injection energy of E_{inj}=10.5 GeV/nucleon, and at the storage energy of E_{coll}=100 GeV/nucleon. What is the change in revolution frequency for particles accelerated from E_{inj} to E_{coll}?

(b) (5 points) If we assume that there are 192 identical dipoles per ring, each of length L = 10 m, what is the required dipole field in each at the collision energy of E_{coll}?

Problem 3. Betatron equations of motion (30 pts)

The betatron equations of motion are

$$x'' - \frac{\rho + x}{\rho^2} = \pm \frac{B_z}{B\rho}\frac{p_0}{p}\left(1 + \frac{x}{\rho}\right)^2, \qquad z'' = \mp \frac{B_x}{B\rho}\frac{p_0}{p}\left(1 + \frac{x}{\rho}\right)^2.$$

Derive these equations for a particle with $p = p_0$ and $q = e$ through the following geometric argument. Let $(\hat{x}, \hat{s}, \hat{z})$ be local polar coordinates inside a dipole (Figure 2). The particle coordinate is

$$\vec{r} = (\rho + x)\,\hat{x} + z\hat{z},$$

where ρ is the bending radius and

$$\hat{x} = -\rho\frac{d\hat{s}}{ds}.$$

In the case of a planar orbit,

$$\hat{x}' = \frac{1}{\rho}\hat{s},$$

where the prime corresponds to the derivative with respect to s. (See SY Lee section 2.I.1 for more details.) The momentum of the particle is $\vec{p} = \gamma m\dot{\vec{r}}$, where γ is constant in the static magnetic field, and the overdot corresponds to the derivative with respect to time t. Similarly, $d\vec{p}/dt = \gamma m\ddot{\vec{r}}$.

(a) (10 pts) Show that

$$\dot{\vec{r}} = \dot{x}\hat{x} + (\rho + x)\,\dot{\theta}\hat{s} + \dot{z}\hat{z},$$

$$\ddot{\vec{r}} = \left[\ddot{x} - (\rho + x)\,\dot{\theta}^2\right]\hat{x} + \left[2\dot{x}\dot{\theta} + (\rho + x)\,\ddot{\theta}\right]\hat{s} + \ddot{z}\hat{z},$$

where $\theta = s/\rho$ is the angle associated with the reference orbit, i.e. $ds = \rho d\theta$.

(b) (10 pts) Using $d\vec{p}/dt = e\vec{v} \times \vec{B}$, with $\vec{B} = B_x\hat{x} + B_z\hat{z}$, show that

$$\ddot{x} - (\rho + x)\,\dot{\theta}^2 = \frac{v_s^2 B_z}{B\rho}, \qquad \ddot{z} = -\frac{v_s^2 B_x}{B\rho},$$

where $B\rho = \gamma m v_s/e$ is the momentum rigidity and v_s is the longitudinal velocity.

(c) (10 pts) Recall that the chain rule for a second derivative is

$$\frac{d^2y}{dx^2} = \frac{d^2y}{du^2}\left(\frac{du}{dx}\right)^2 + \frac{dy}{du}\frac{d^2u}{dx^2}$$

and assume $d^2s/dt^2 = 0$. Transform the time coordinate to the longitudinal distance s with $ds = \rho d\theta$, where $d\theta = v_s dt/\left(\rho + x\right)$, and show that

$$x'' - \frac{\rho + x}{\rho^2} = \frac{B_z}{B\rho}\left(1 + \frac{x}{\rho}\right)^2, \qquad z'' = -\frac{B_x}{B\rho}\left(1 + \frac{x}{\rho}\right)^2.$$

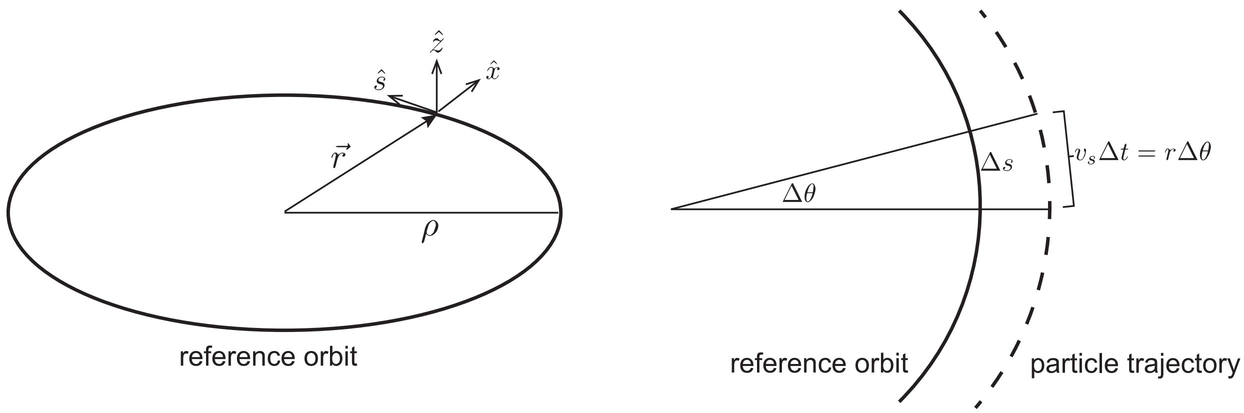

Figure 2: Coordinate system.